LECTURE PRESENTATION

System Analysis & Design

System Interface Design, Navigation Design, Input Design and Output Design

Press → to begin · F for fullscreen

INTRODUCTION

Learning Overview

Seven parts — from the principles behind good interfaces to applying them in a real system.

01 **UI Design Principles**
The foundations of a usable interface

02 **UI Design Process**
From use scenarios to evaluation

03 **Navigation Design**
Helping users move through the system

04 **Input Design**
Capturing accurate data efficiently

05 **Output Design**
Presenting information that matters

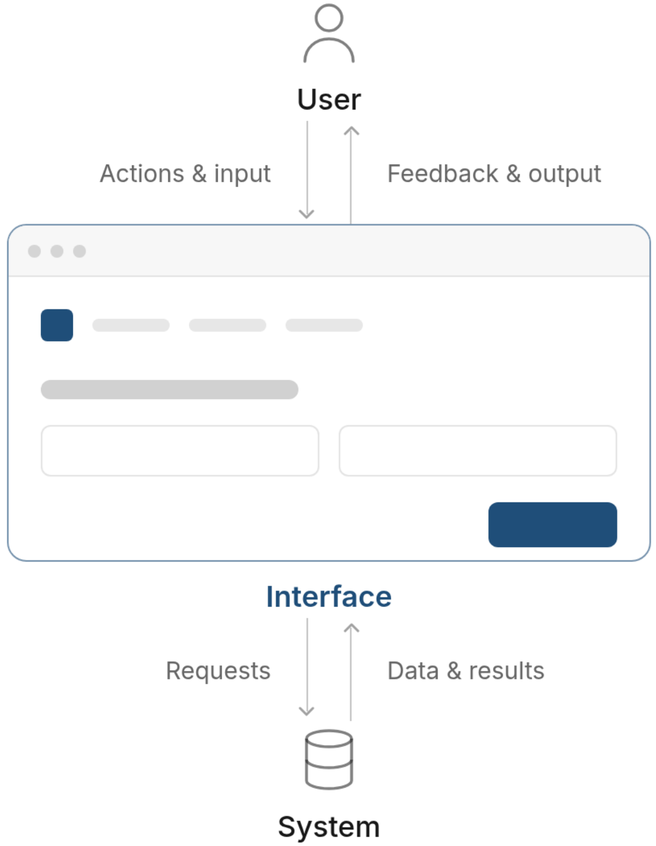
06 **Tune Source Case Study**
Applying every concept to one system

07 **Summary**
Key takeaways and discussion

What is User Interface Design?

The interface is the part of the system users actually see — the screens, forms and reports through which they get their work done.

- **User interaction**
  How people communicate with the system
- **User satisfaction**
  How people feel while using it
- **System efficiency**
  How quickly and reliably tasks get done
- **Usability**
  How easy the system is to learn and use
- **Accessibility**
  Whether everyone is able to use it



Core UI Principles

01

Layout

Organise every screen into predictable, consistent areas.

02

Content Awareness

Users always know where they are and what they see.

03

Aesthetics

Pleasing, functional visuals with minimal clutter.

04

User Experience

Easy to learn, easy to use, satisfying to work with.

05

Consistency

The same things look and behave the same way.

06

Minimize User Effort

Fewer clicks, less typing, less to remember.

07

Accessibility

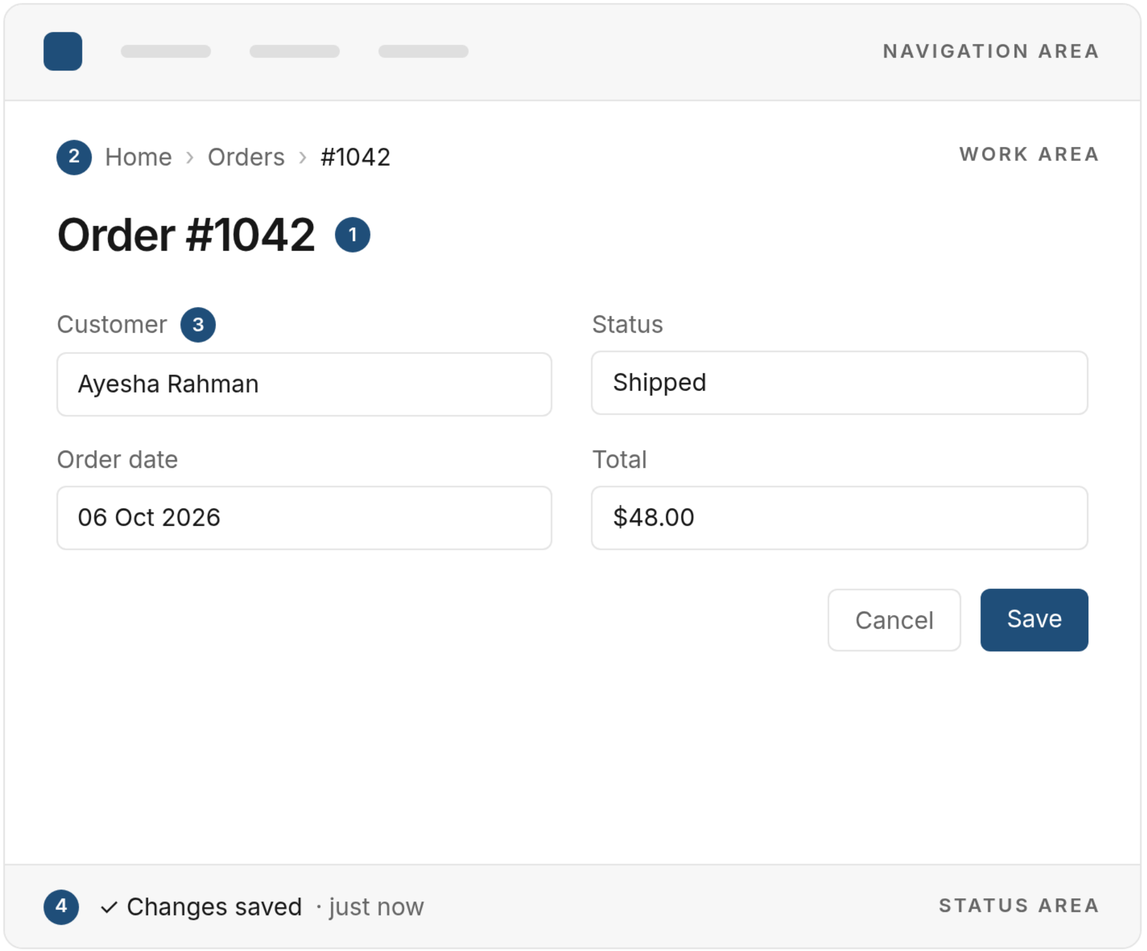
Usable by people of all abilities and devices.

08

Scalability

Designs that grow with new features and users.

Layout & Content Awareness



Layout

Divide each screen into navigation, work and status areas — and keep them in the same place on every screen.

Content awareness

Users should always know where they are and what the system is doing.

1. **Page titles** — Name the current screen
2. **Breadcrumbs** — Show the path — and a way back
3. **Labels** — Describe every field and area
4. **System feedback** — Confirm what just happened

Aesthetics & User Experience

Aesthetics

How the interface looks.



Colors

A limited palette with strong contrast; colour supports meaning but never carries it alone.

Fonts

One or two readable typefaces with a clear, consistent size hierarchy.

Whitespace

Group related items and give content room to breathe — avoid dense screens.

User Experience

How the interface feels to use.



Responsiveness

Immediate feedback on every action, and layouts that adapt to any screen size.

Error handling

Prevent errors first; when they happen, explain clearly how to recover.

Accessibility

Keyboard support, screen-reader labels and sufficient colour contrast.

Consistency & Minimize User Effort

CONSISTENCY

The same things look and behave the same way.

Internal

Within the system — same layout, terms, colours and button positions on every screen.

> “Save” is always bottom-right

External

With other systems and platform conventions users already know.

> Ctrl + S saves · logo returns home

MINIMIZE USER EFFORT

Fewer clicks, less typing, less to remember.

Autofill

Pre-fill data the system already knows

Predictive text

Suggest values as the user types

Shortcuts

Keyboard accelerators for frequent actions

Remember me

Keep returning users signed in

Three-clicks rule: users should be able to reach any function or information in no more than three clicks.

UI Design Process Overview

Five steps, with prototyping and evaluation repeated until the design works for users.



Use Scenario Development

Describe how users will work with the system

Interface Structure Design

Define the screens and how they connect

Interface Standards Design

Agree on metaphors, objects, actions and templates

Interface Prototyping

Build mock-ups of screens, forms and reports

Interface Evaluation

Test with users and refine the design

Use Scenario Development

A use scenario outlines the steps a user performs to complete one piece of their work.

- **User goals**
  What the user is trying to achieve
- **Workflows**
  The sequence of steps they follow
- **Edge cases**
  Unusual but valid situations
- **Error handling**
  What happens when something goes wrong

EXAMPLE SCENARIO — E-commerce

Buying headphones online

User — Returning customer, signed in
Goal — Buy a pair of wireless headphones under $80

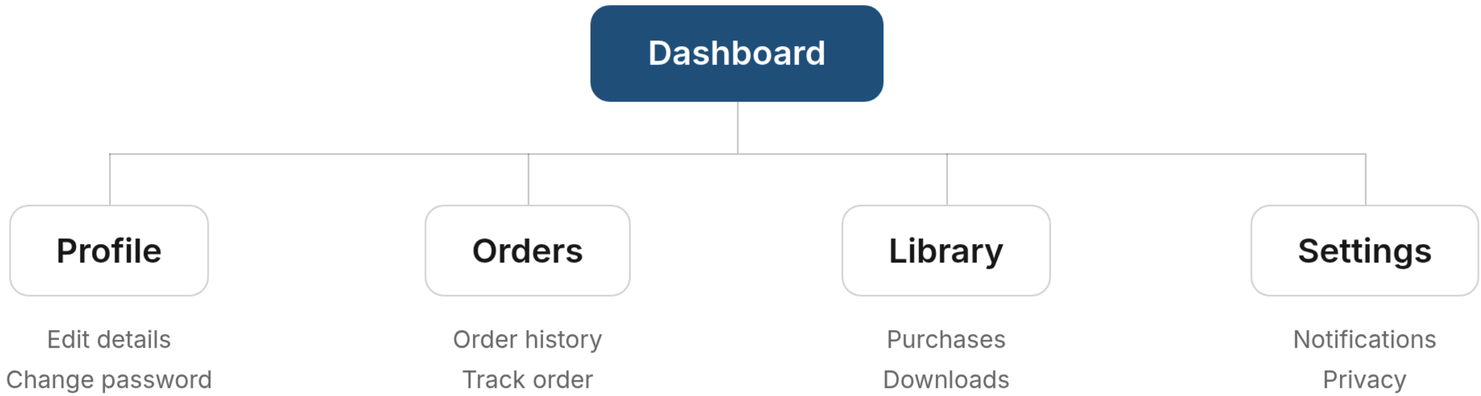
1. Searches for “wireless headphones”
2. Filters by price and compares two products
3. Adds one item to the cart
4. Confirms saved address and pays by card
5. Receives an order confirmation by email

EDGE CASE
Item goes out of stock → suggest similar products

ERROR
Card declined → keep the cart, ask for another card

Interface Structure Design

An interface structure diagram shows every screen and how users move between them.



Screen relationships

Which screens exist and how they connect

User journey

The path a user takes to finish a task

Navigation flow

How users move forward, back and across

Interface Standards & Prototyping

INTERFACE STANDARDS

The basic design elements shared by every screen in the system.

Interface metaphor

A real-world concept that guides the design — a shopping cart, a desktop, a folder.

Objects, actions & icons

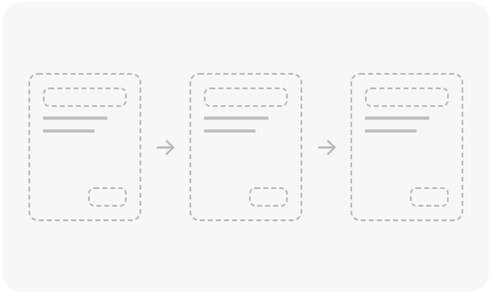
The same names, verbs and symbols used everywhere.

Templates

A standard layout reused across screens.

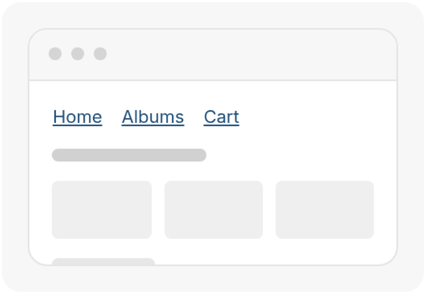
INTERFACE PROTOTYPING

A mock-up of screens, forms or reports — moving from rough sketch to near-final design.



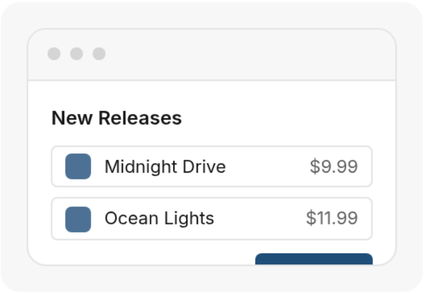
Storyboard

Paper sketches of screens and how they link



HTML Prototype

Web pages with working links but no real data



Language Prototype

Built in the final language — looks real, partly functional

Low fidelity · fast and cheap to change — High fidelity · realistic, slower to change

Interface Evaluation

01

Heuristic Evaluation

Experts check the prototype against a checklist of design principles.

USER INVOLVEMENT

None — experts only

02

Walk-through Evaluation

The team walks users through the prototype and gathers their reactions.

USER INVOLVEMENT

Users watch

03

Interactive Evaluation

Users work with the prototype themselves while an analyst observes.

USER INVOLVEMENT

Users try it

04

Formal Usability Testing

Users perform real tasks in a controlled setting; actions are recorded.

USER INVOLVEMENT

Users are measured

Quicker · cheaper → More realistic · more costly

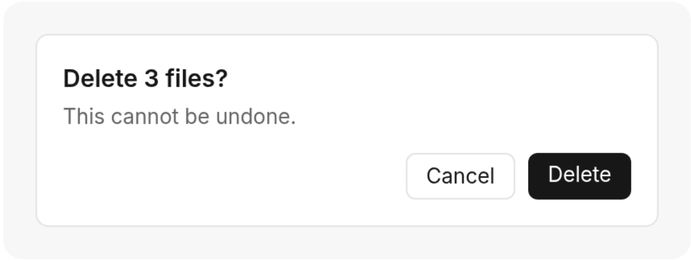
Navigation Design Principles

Navigation should be so simple that users don’t have to think about it.

01

Prevent mistakes

Label commands clearly, limit choices, and ask before destructive actions.



02

Simplify recovery

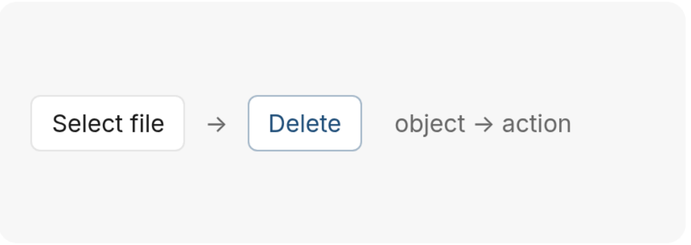
Make actions reversible — undo, back and cancel should always be close.



03

Consistent grammar order

Use one command order everywhere: object then action, or action then object.



Navigation Controls

Language Commands

Users type commands or natural-language requests.



STRENGTH

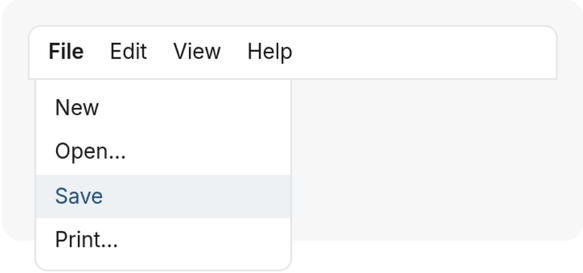
Fast and powerful for experts

LIMITATION

Users must remember the syntax

Menus

Users choose from a list of available options.



STRENGTH

Easy to learn — nothing to memorise

LIMITATION

Slower for frequent, repeated tasks

Direct Manipulation

Users act on objects directly — drag, drop, resize.



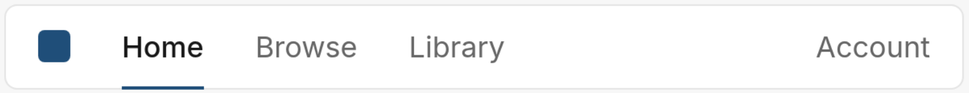
STRENGTH

Intuitive, with immediate feedback

LIMITATION

Awkward for complex or bulk actions

Common Navigation Components



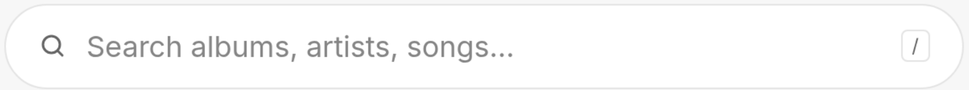
Menu Bar

Primary, always-visible navigation across the top of every screen.



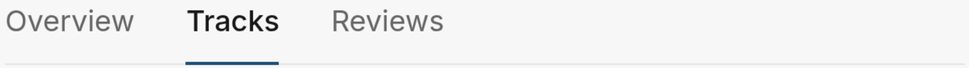
Breadcrumbs

Show where the user is in a hierarchy and offer a one-click way back.



Search Bar

Jump straight to content in large systems instead of browsing.



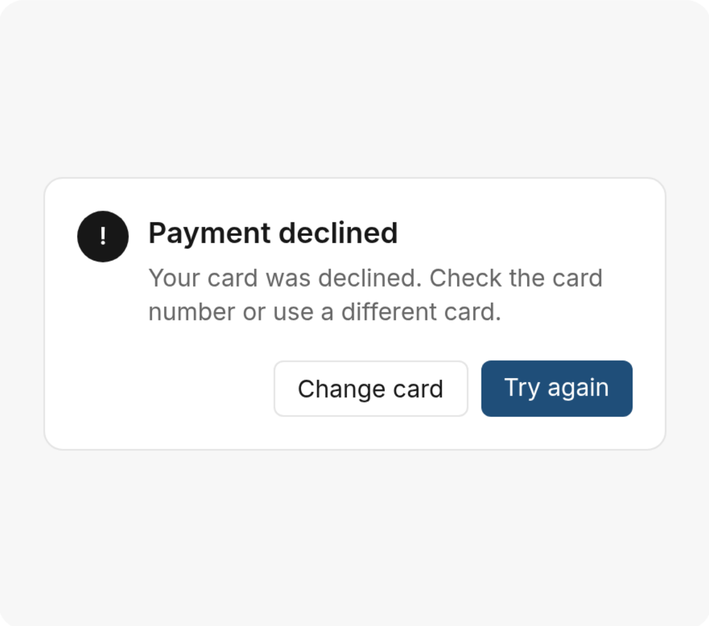
Tabs

Switch between related views of the same item without leaving the screen.

Menu Types

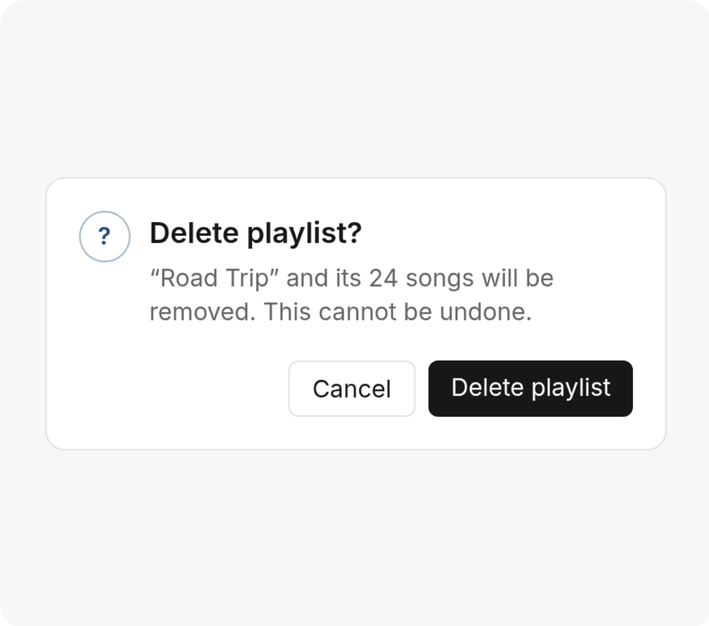
MENU TYPE	WHAT IT IS	WHEN TO USE
Menu Bar	List of commands across the top of the screen	Main menu of the system — always visible
Drop-down Menu	Menu that drops down below a menu-bar item	Second-level options grouped under the menu bar
Hyperlink Menu	List of links to other pages	Main menu of a web-based system
Embedded Hyperlinks	Links placed within body text	Pointing to related content in context
Pop-up Menu	Appears at the cursor on right-click or long-press	Shortcuts for experienced users
Tab Menu	Multi-page view with a tab for each page	Switching between related forms or settings
Image Map	Clickable regions within an image	Only when the image itself adds meaning — maps, seat plans

Message Design



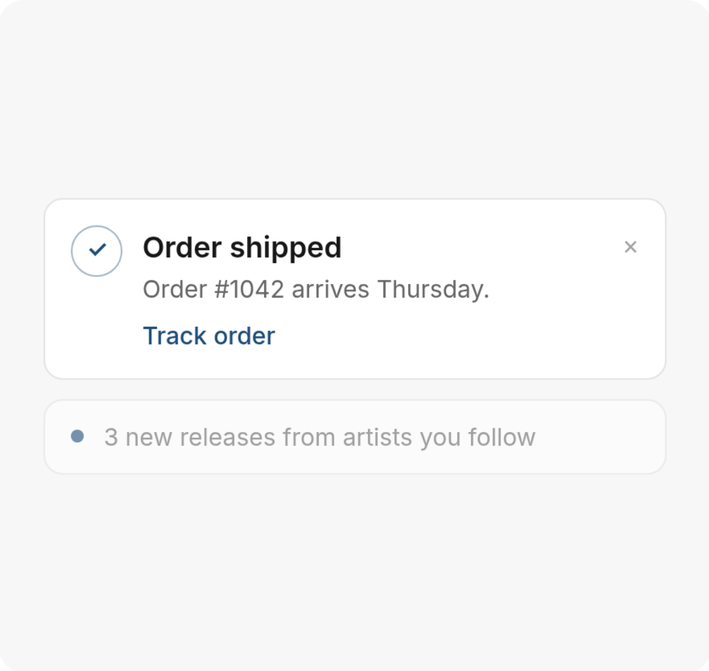
Error messages

Say what went wrong and how to fix it — never blame the user.



Confirmation messages

Ask before irreversible actions; name the action on the button.



Notification messages

Inform without interrupting the user’s current task.

EVERY MESSAGE Clear, concise and consistent — in the user’s language, free of technical jargon.

Input Design Overview

Goal: capture accurate information for the system — simply and easily.

01

Data accuracy

Capture correct data the first time it is entered.

02

Data integrity

Keep data valid and consistent across the whole system.

03

User satisfaction

Make data entry quick, clear and painless.

04

Efficient data entry

Fewer keystrokes, fewer screens, less re-typing.

THREE INPUT PRINCIPLES

Use online processing when possible, for immediate feedback

Capture data at the source to avoid copying and re-keying

Minimise keystrokes with defaults, lists and automation

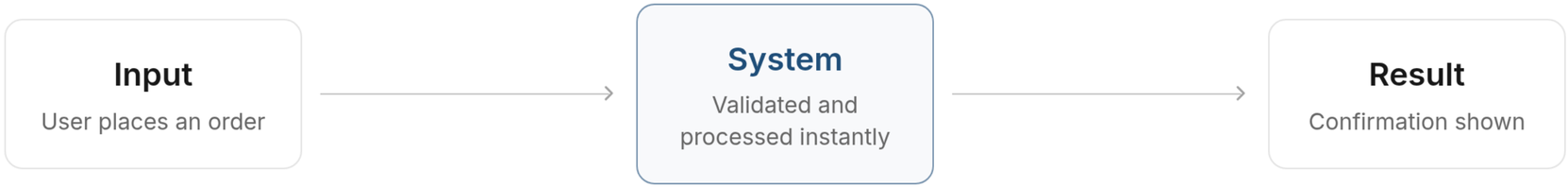
Online vs Batch Processing

Online

Immediate

Each transaction is processed the moment it is entered.

e.g. ATM withdrawal, flight booking, online checkout

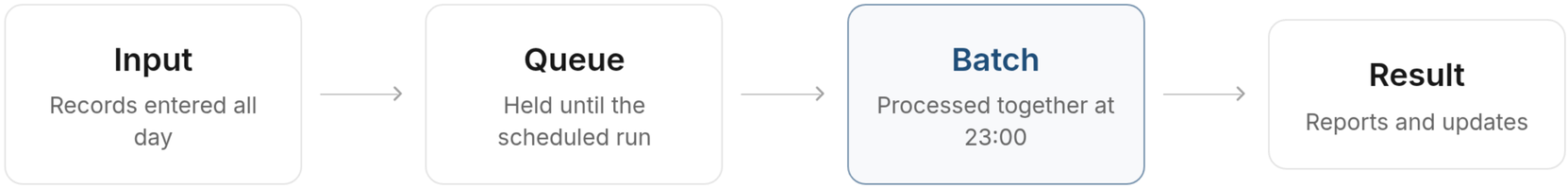


Batch

Scheduled

Transactions are collected and processed together at set times.

e.g. payroll, monthly billing, end-of-day bank updates



Prefer online processing where possible — errors are caught while the user is still there to fix them.

Data Capture Technologies

Source data automation: capture data electronically, where it originates.

Barcode Reader

Scans printed codes to identify items instantly.

Used in · Retail checkout, inventory

OCR

Converts printed or handwritten text into digital data.

Used in · Forms, cheques, ID cards

Magnetic Stripe

Reads data stored on a magnetic strip.

Used in · Bank cards, hotel keys

Smart Card

An embedded chip stores and processes data securely.

Used in · Bank cards, SIM cards, IDs

RFID

Radio tags are read without contact or line of sight.

Used in · Toll gates, library books

Web Forms

Users enter data directly into the system online.

Used in · Registration, checkout

Input Controls

Full name

e.g. Ayesha Rahman

Text Box

Best for · Free-form data such as names, addresses and comments.

Genre

Jazz

Dropdown

Best for · Pick one value from a long list while saving space.

Notify me about

- [x] New releases
- [ ] Price drops
- [x] Concerts nearby

Checkbox

Best for · Select any number of independent yes/no options.

Delivery

- (•) Standard (3–5 days)
- ( ) Express (next day)
- ( ) Store pickup

Radio Button

Best for · Choose exactly one from a few visible options.

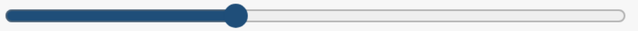
Artist

Type or choose...

Combo Box

Best for · Type a value or choose from a list — ideal for long lists.

Maximum price $40



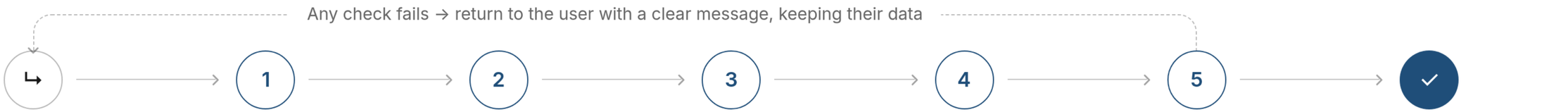
$5 $100

Slider

Best for · Choose from a continuous range when exact precision isn't needed.

Input Validation

Every input passes through a series of checks before it is accepted.



User Input
Data is entered on a form

Form submitted

Completeness
All required fields are filled

“Name is required”

Format
Right type and pattern

“Phone has letters”

Range
Values within allowed limits

“Quantity 1–10”

Consistency
Related fields agree

“End before start”

Database
Values match stored records

“Customer ID not found”

Accepted
Saved to the system

Also common: check-digit validation for codes such as ISBNs and card numbers.

Validation Examples

Book a stay

Reset example

Email

ayesha.rahman@mail

× Enter a valid email, e.g. name@example.com

Phone number

0171-234567

× Must be 11 digits — 10 entered

Card expiry (MM/YY)

03/24

× This card expired in 03/24

Date of birth

31/02/2004

× Not a real date — use DD/MM/YYYY

Check-in

12/10/2026

✓ Looks good

Check-out

10/10/2026

× Check-out must be after check-in

RULES BEHIND THE FORM

Edit the fields — messages update as you type.

Email
One “@”, a domain and an ending such as .com
Format

Phone number
Digits only, exactly 11 digits
Format Range

Card expiry
Valid MM/YY that is not in the past
Format Range

Date
A real calendar date — no 31 February
Format

Cross-field
Check-out must be after check-in
Consistency

Output Design Principles

Outputs are what most users see of a system — reports, screens and messages.

01

Relevance

Include only the information users need for their task.

ASK
Who reads this, and for what decision?

02

Accuracy

Correct, complete and verifiable data.

ASK
Can users trust every number?

03

Timeliness

Delivered when it is needed — early enough to act on.

ASK
When is this needed?

04

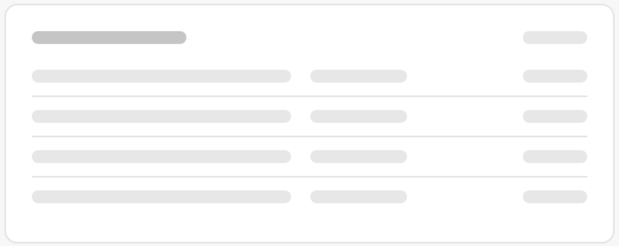
Accessibility

Readable and usable by everyone, on every medium.

ASK
Can every user read and use it?

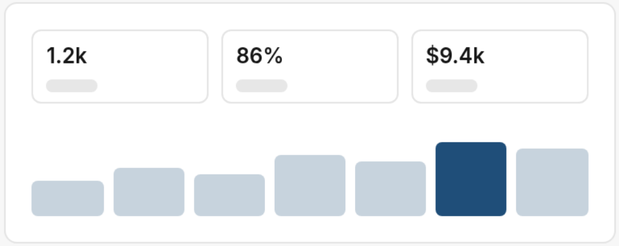
Also: understand how the output will be used, manage information load, and minimise bias in how data is presented.

Types of Outputs



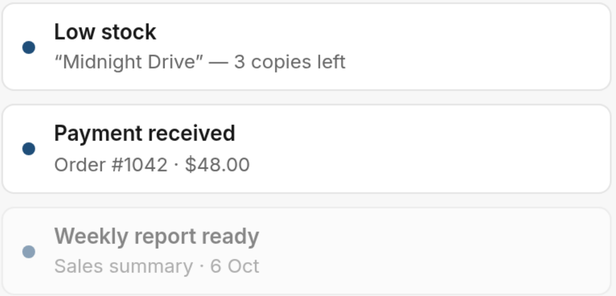
Reports

Structured documents — detail, summary, exception and turnaround reports.



Dashboards

An at-a-glance view of key measures, often updated in real time.



Notifications

Short messages that prompt action — email, SMS, push, in-app.

OUTPUT MEDIA

Print	Web	Mobile	PDF	Excel
Formal records	Interactive, current	On the go	Fixed layout to share	Further analysis

Tune Source Music Store

Case scenario: Modernising an independent music retailer with web and in-store digital systems.

Tune Source needs an intuitive omnichannel interface allowing music enthusiasts to discover artists, listen to sample tracks, order physical vinyl/CDs, or download lossless digital albums.

Digital & Physical Catalog

40,000+ ALBUMS

Unified browsing for vinyl, CDs, high-res FLAC downloads, and streaming previews.

Customer Experience Center

PERSONALIZED PORTALS

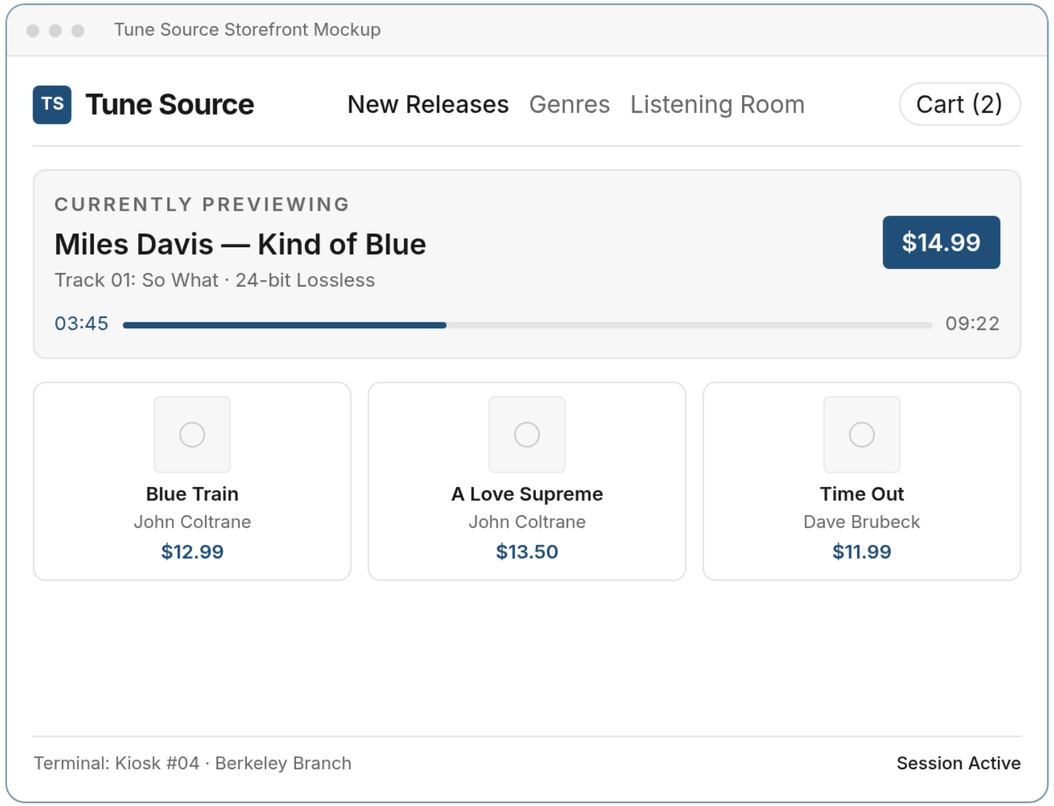
Customer profiles, saved payment tokens, wishlist, and instant download vault.

Omnichannel Checkout

WEB & KIOSK TERMINALS

Seamless transaction pipeline for both web users and in-store interactive listening kiosks.

Analytical focus: Seamless transition between discovery, media sampling, and verified transactional checkout.



Applying Concepts at Tune Source

How each lecture domain translates directly into architectural decisions for the system.

01 — Principle

UI Principles

Consistency & Effort

APPLICATION

3-click music discovery, sticky mini-player bar, standardized visual hierarchy.

OUTCOME

Low cognitive load during exploration

02 — Structure

Navigation

Taxonomy & Breadcrumbs

APPLICATION

Genre tree (Genre › Artist › Album), persistent search with auto-suggest query filter.

OUTCOME

Users never get lost in deep catalog

03 — Capture

Input Design

Optimized Controls

APPLICATION

Dropdowns for format selection (Vinyl/CD/FLAC), stepper for quantity, saved address selection.

OUTCOME

Prevents typing errors & friction

04 — Integrity

Validation

Real-Time Pipeline

APPLICATION

Format checks for card number/expiry, inventory verification prior to payment submission.

OUTCOME

Zero bad orders sent to fulfillment

05 — Presentation

Output Design

Relevance & Formats

APPLICATION

Instant on-screen receipt, downloadable PDF invoice, digital license key via secure portal.

OUTCOME

Immediate verification & trust

System Synergy: Strong navigation gets users to the item; streamlined input and validation ensures seamless purchase; clear output builds customer confidence.

Tune Source User Journey

End-to-end task analysis: tracing a customer transaction from exploration to completion.

1 → 2 → 3 → 4 → 5

1 Browse Music

Filter by genre or search by artist

INTERFACE CONTROLS

Search bar, category tabs, filter chips

SYSTEM FEEDBACK

Dynamic live search results

2 View Album

Audition 30s clips, choose Vinyl or FLAC

INTERFACE CONTROLS

Audio player, format selector dropdown

SYSTEM FEEDBACK

High-res artwork & track duration

3 Add To Cart

Confirm format, quantity & shipping preference

INTERFACE CONTROLS

Primary CTA button, quantity stepper

SYSTEM FEEDBACK

Animated badge count & toast confirmation

4 Checkout

Provide shipping address and payment details

INTERFACE CONTROLS

Standardized forms with auto-fill

SYSTEM FEEDBACK

Inline validation on every field

5 Confirmation

Review invoice, download files, view tracking

INTERFACE CONTROLS

Print receipt, token download button

SYSTEM FEEDBACK

Order #TS-4029 issued immediately

Analytical Takeaway — Human-Computer Interaction Principle

Every stage pairs user intent with immediate system feedback. When error rates spike at step 4 (Checkout), UI analysts re-evaluate input form complexity and validation messages.

07 — SUMMARY

Activity & Discussion

Seminar debate: Evaluating the impact of interface consistency on large-scale software systems.

Q1 DISCUSSION PROMPT 1

How does consistency improve user experience?

Consider cognitive workload, learning curves, and user confidence when interacting with unfamiliar modules.

GUIDING ANGLES

- **Cognitive load & familiarity**
  Users transfer mental models from prior systems without relearning.
- **Error prevention**
  Standardized layouts and button placement avoid accidental destructive actions.
- **Speed & muscle memory**
  Predictable navigation allows power users to complete frequent tasks rapidly.

Q2 DISCUSSION PROMPT 2

What strategies help maintain consistency across a system?

How can multi-disciplinary engineering and design teams ensure coherence as a system scales over time?

PRACTICAL STRATEGIES

- **Design systems & UI token libraries**
  Centralize color tokens, typography scales, spacing units, and components.
- **Interface standards documentation**
  Formalize metaphors, terminology, screen templates, and icon meanings.
- **Heuristic evaluation & peer audits**
  Systematically inspect new screens against standards before deployment.

LECTURE CONCLUSION

Key Takeaways

01

UI Principles

Balance layout, aesthetics, consistency, and effort minimization for high usability.

02

UI Process

Iterate systematically from use scenarios through structure to prototyping and evaluation.

03

Navigation Design

Structure hierarchies, provide breadcrumbs, and ensure actions are easily reversible.

04

Input Design

Capture data at source; choose the right control; minimize manual user typing.

05

Validation

Enforce multi-tier validation (format, range, consistency, database) before accepting data.

06

Output Design

Deliver relevant, accurate, timely information across appropriate media and formats.

Thank You.

Questions & Academic Discussion

System Analysis & Design

Department of Computer Science & Engineering

- Course Lecture Presentation · Final Slide